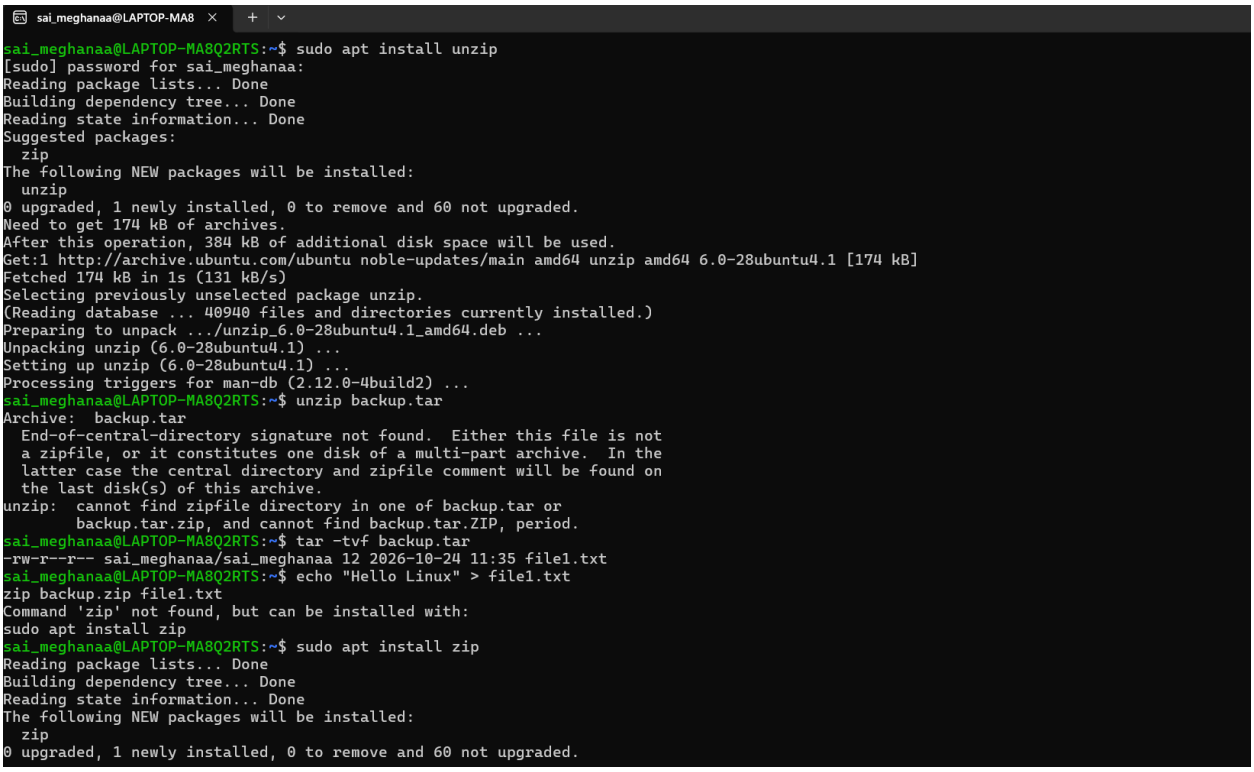
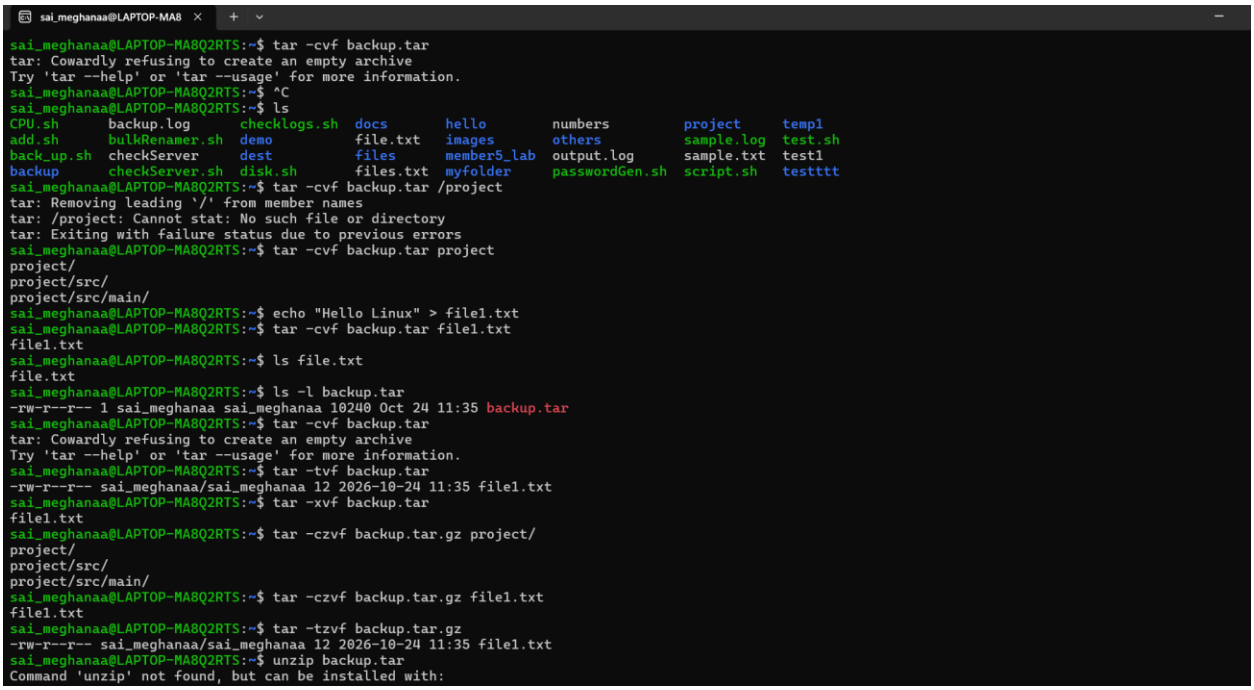
# Linux Basic Commands:



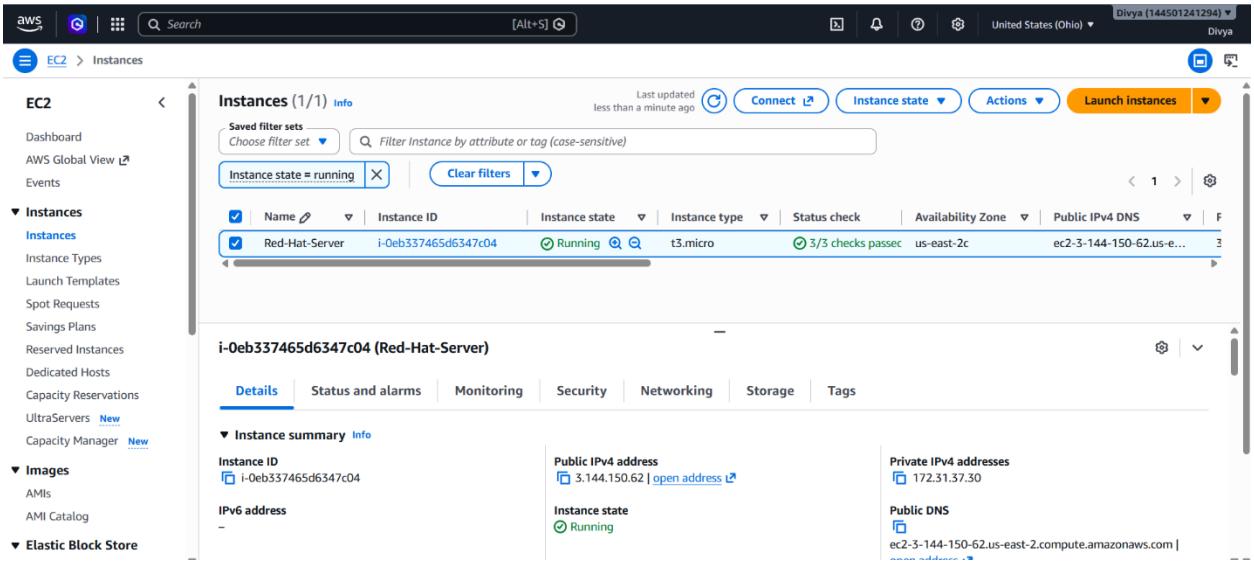
```
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ sudo apt install unzip
[sudo] password for sai_meghanaa:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  zip
The following NEW packages will be installed:
  unzip
0 upgraded, 1 newly installed, 0 to remove and 60 not upgraded.
Need to get 174 kB of archives.
After this operation, 384 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 unzip amd64 6.0-28ubuntu4.1 [174 kB]
Fetched 174 kB in 1s (131 kB/s)
Selecting previously unselected package unzip.
(Reading database ... 40940 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-28ubuntu4.1_amd64.deb ...
Unpacking unzip (6.0-28ubuntu4.1) ...
Setting up unzip (6.0-28ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ unzip backup.tar
Archive:  backup.tar
  End-of-central-directory signature not found.  Either this file is not
  a zipfile, or it constitutes one disk of a multi-part archive.  In the
  latter case the central directory and zipfile comment will be found on
  the last disk(s) of this archive.
unzip:  cannot find zipfile directory in one of backup.tar or
        backup.tar.zip, and cannot find backup.tar.ZIP, period.
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -tvf backup.tar
-rw-r--r-- sai_meghanaa/sai_meghanaa 12 2026-10-24 11:35 file1.txt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ echo "Hello Linux" > file1.txt
zip backup.zip file1.txt
Command 'zip' not found, but can be installed with:
sudo apt install zip
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ sudo apt install zip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  zip
0 upgraded, 1 newly installed, 0 to remove and 60 not upgraded.
```

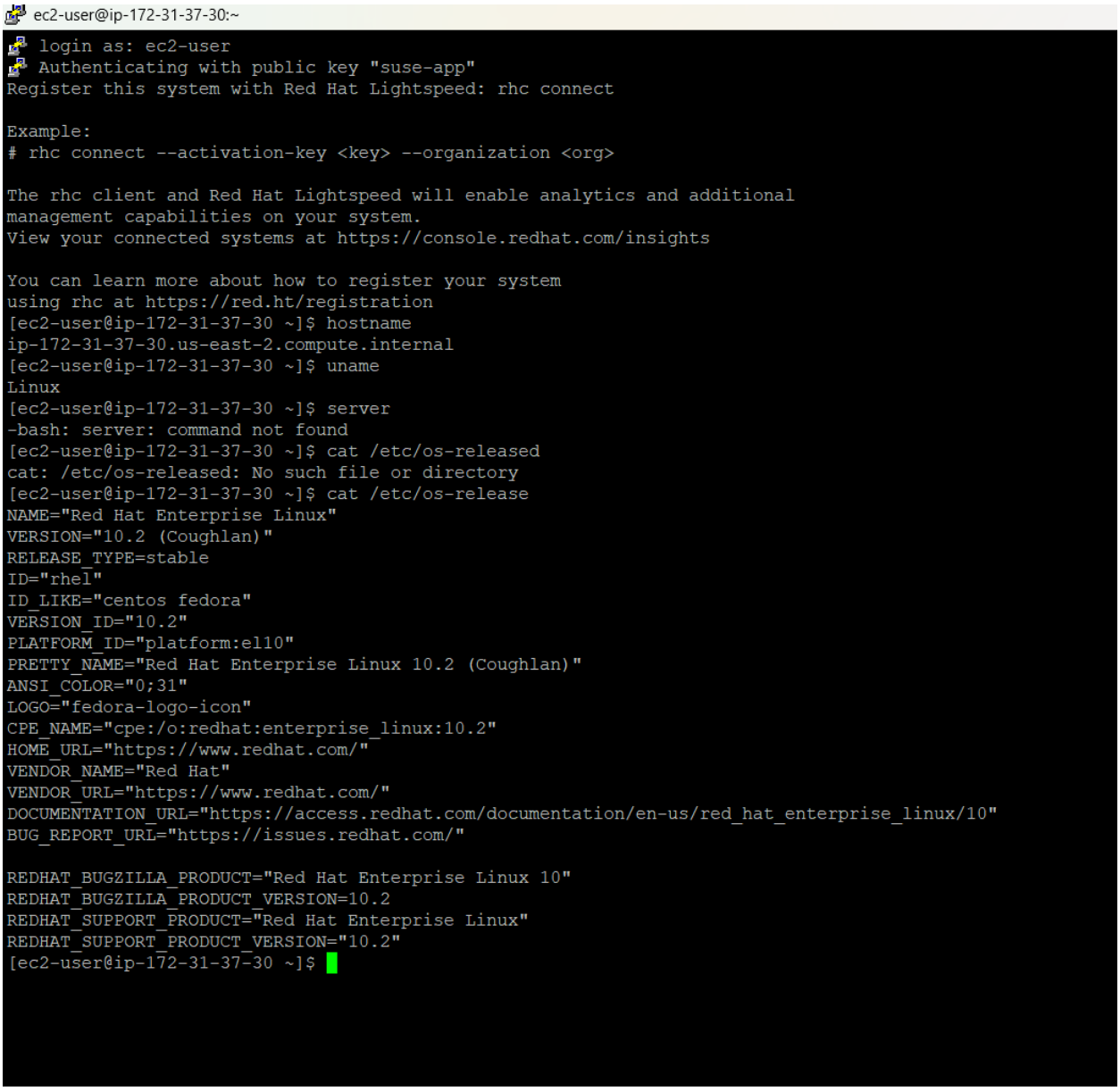


```
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -cvf backup.tar
tar: Cowardly refusing to create an empty archive
Try 'tar --help' or 'tar --usage' for more information.
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ ^C
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ ls
CPU.sh      backup.log      checklogs.sh  docs       hello        numbers         project     temp1
add.sh      bulkRenamer.sh  demo          file.txt   images       others          sample.log  test.sh
back_up.sh  checkServer     dest          files      member5_lab  output.log      sample.txt  test1
backup      checkServer.sh  disk.sh       files.txt  myfolder     passwordGen.sh  script.sh   testttt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -cvf backup.tar /project
tar: Removing leading `/' from member names
tar: /project: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -cvf backup.tar project
project/
project/src/
project/src/main/
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ echo "Hello Linux" > file1.txt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -cvf backup.tar file1.txt
file1.txt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ ls file.txt
file.txt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ ls -l backup.tar
-rw-r--r-- 1 sai_meghanaa sai_meghanaa 10240 Oct 24 11:35 backup.tar
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -cvf backup.tar
tar: Cowardly refusing to create an empty archive
Try 'tar --help' or 'tar --usage' for more information.
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -tvf backup.tar
-rw-r--r-- sai_meghanaa/sai_meghanaa 12 2026-10-24 11:35 file1.txt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -xvf backup.tar
file1.txt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -czvf backup.tar.gz project/
project/
project/src/
project/src/main/
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -czvf backup.tar.gz file1.txt
file1.txt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -tzvf backup.tar.gz
-rw-r--r-- sai_meghanaa/sai_meghanaa 12 2026-10-24 11:35 file1.txt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ unzip backup.tar
Command 'unzip' not found, but can be installed with:
```

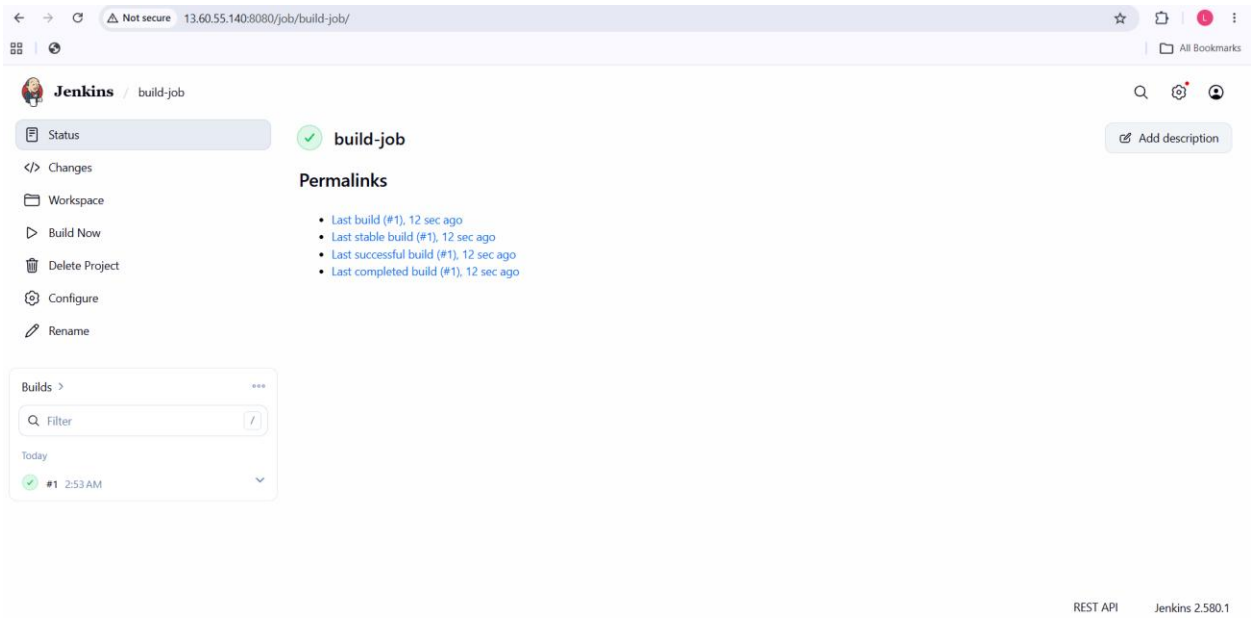
```
sai_meghanaa@LAPTOP-MA8
sudo apt install zip
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ sudo apt install zip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  zip
0 upgraded, 1 newly installed, 0 to remove and 60 not upgraded.
Need to get 176 kB of archives.
After this operation, 549 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 zip amd64 3.0-13ubuntu0.2 [176 kB]
Fetched 176 kB in 2s (91.1 kB/s)
Selecting previously unselected package zip.
(Reading database ... 40958 files and directories currently installed.)
Preparing to unpack .../zip_3.0-13ubuntu0.2_amd64.deb ...
Unpacking zip (3.0-13ubuntu0.2) ...
Setting up zip (3.0-13ubuntu0.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ zip backup.zip file1.txt
  adding: file1.txt (stored 0%)
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ echo "Hello Linux" > file1.txt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ zip backup.zip file1.txt
updating: file1.txt (stored 0%)
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ unzip backup.zip
Archive:  backup.zip
replace file1.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
 extracting: file1.txt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ tar -tvf backup.tar
-rw-r--r-- sai_meghanaa/sai_meghanaa 12 2026-10-24 11:35 file1.txt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$
```

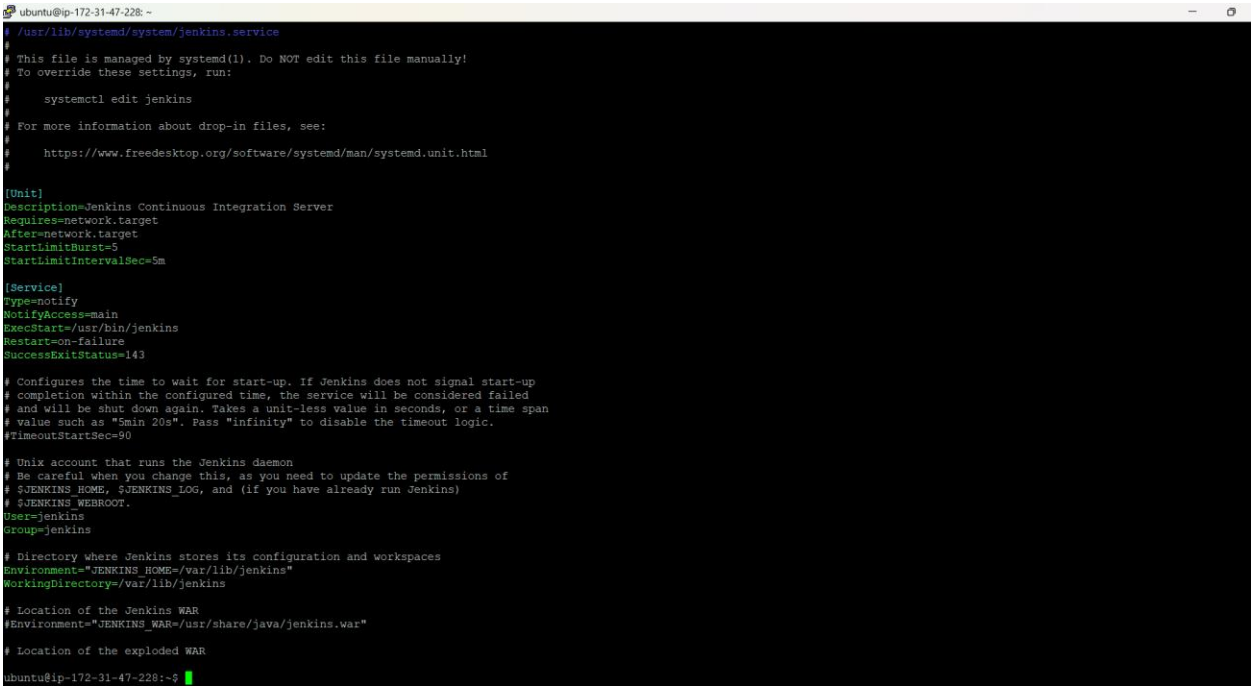
# Install RedHat OS:

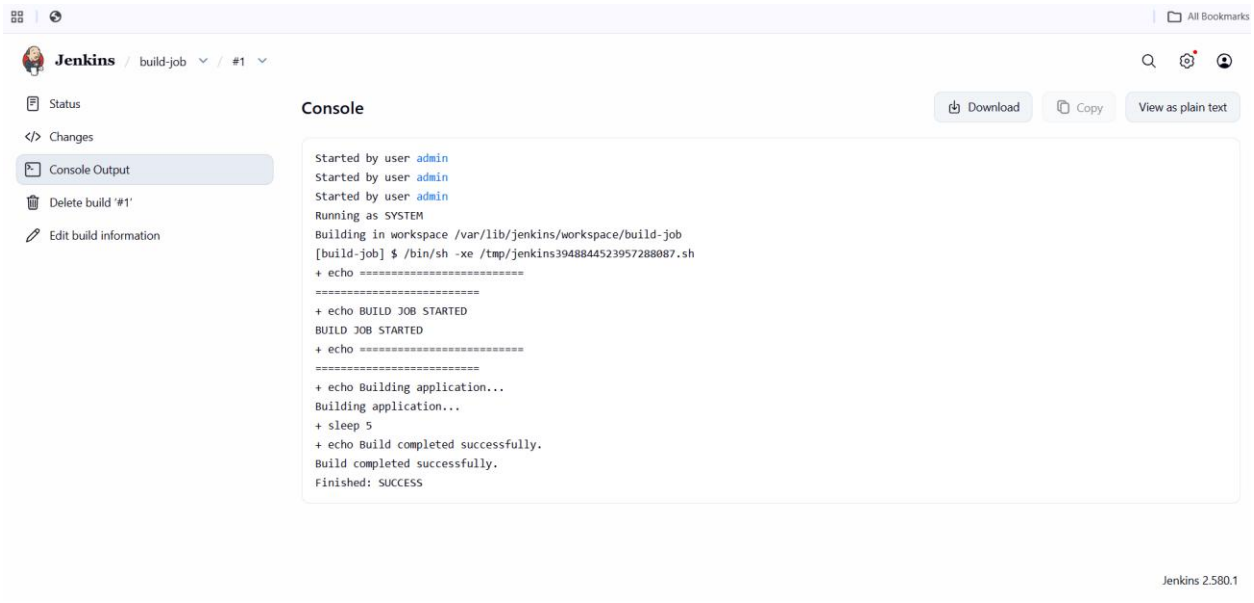



ec2-user@ip-172-31-37-30:~
login as: ec2-user
Authenticating with public key "suse-app"
Register this system with Red Hat Lightspeed: rhc connect
Example:
# rhc connect --activation-key <key> --organization <org>
The rhc client and Red Hat Lightspeed will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights
You can learn more about how to register your system
using rhc at https://red.ht/registration
[ec2-user@ip-172-31-37-30 ~]$ hostname
ip-172-31-37-30.us-east-2.compute.internal
[ec2-user@ip-172-31-37-30 ~]$ uname
Linux
[ec2-user@ip-172-31-37-30 ~]$ server
-bash: server: command not found
[ec2-user@ip-172-31-37-30 ~]$ cat /etc/os-released
cat: /etc/os-released: No such file or directory
[ec2-user@ip-172-31-37-30 ~]$ cat /etc/os-release
NAME="Red Hat Enterprise Linux"
VERSION="10.2 (Coughlan)"
RELEASE_TYPE=stable
ID="rhel"
ID_LIKE="centos fedora"
VERSION_ID="10.2"
PLATFORM_ID="platform:el10"
PRETTY_NAME="Red Hat Enterprise Linux 10.2 (Coughlan)"
ANSI_COLOR="0;31"
LOGO="fedora-logo-icon"
CPE_NAME="cpe:/o:redhat:enterprise_linux:10.2"
HOME_URL="https://www.redhat.com/"
VENDOR_NAME="Red Hat"
VENDOR_URL="https://www.redhat.com/"
DOCUMENTATION_URL="https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/10"
BUG_REPORT_URL="https://issues.redhat.com/"
REDHAT_BUGZILLA_PRODUCT="Red Hat Enterprise Linux 10"
REDHAT_BUGZILLA_PRODUCT_VERSION=10.2
REDHAT_SUPPORT_PRODUCT="Red Hat Enterprise Linux"
REDHAT_SUPPORT_PRODUCT_VERSION="10.2"
[ec2-user@ip-172-31-37-30 ~]$

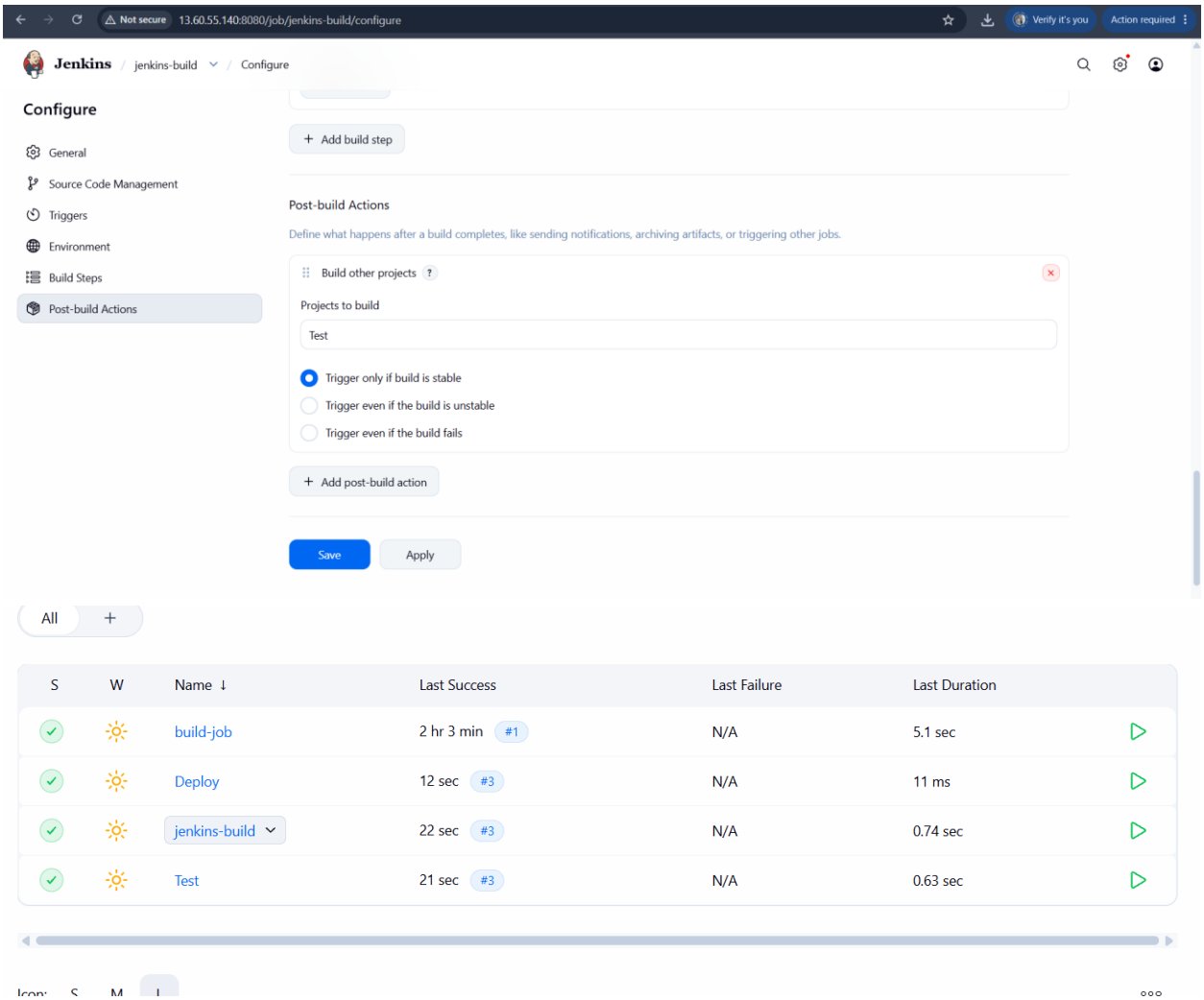
# Jenkins Job Creation, Ordering, and Scheduling:



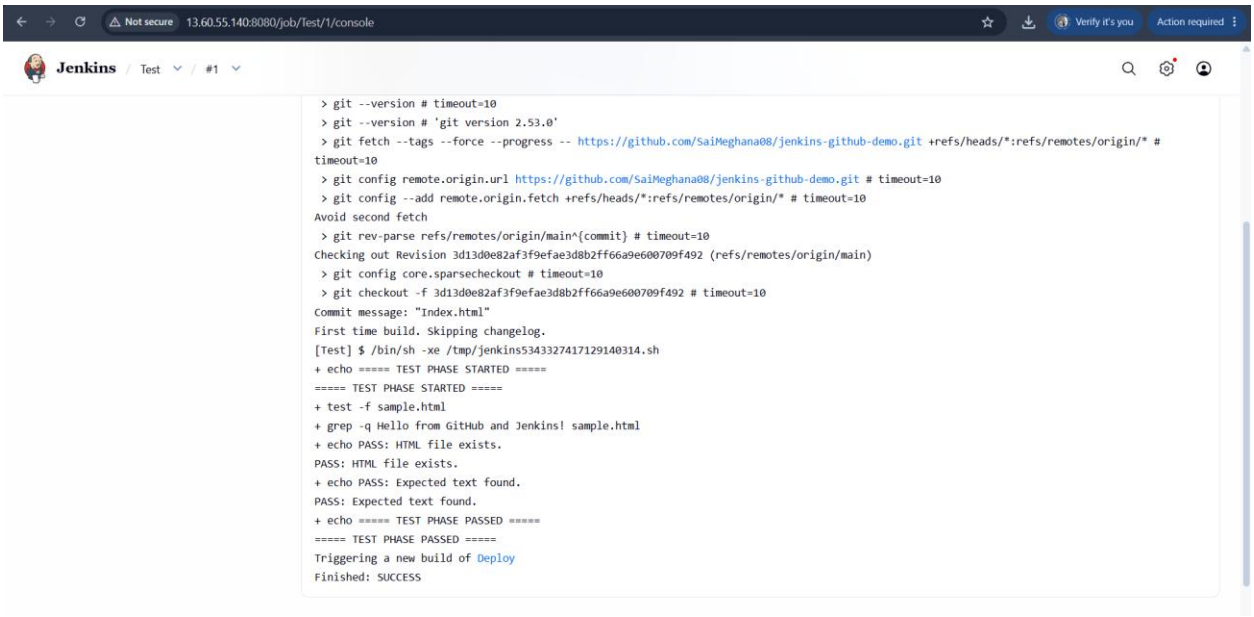



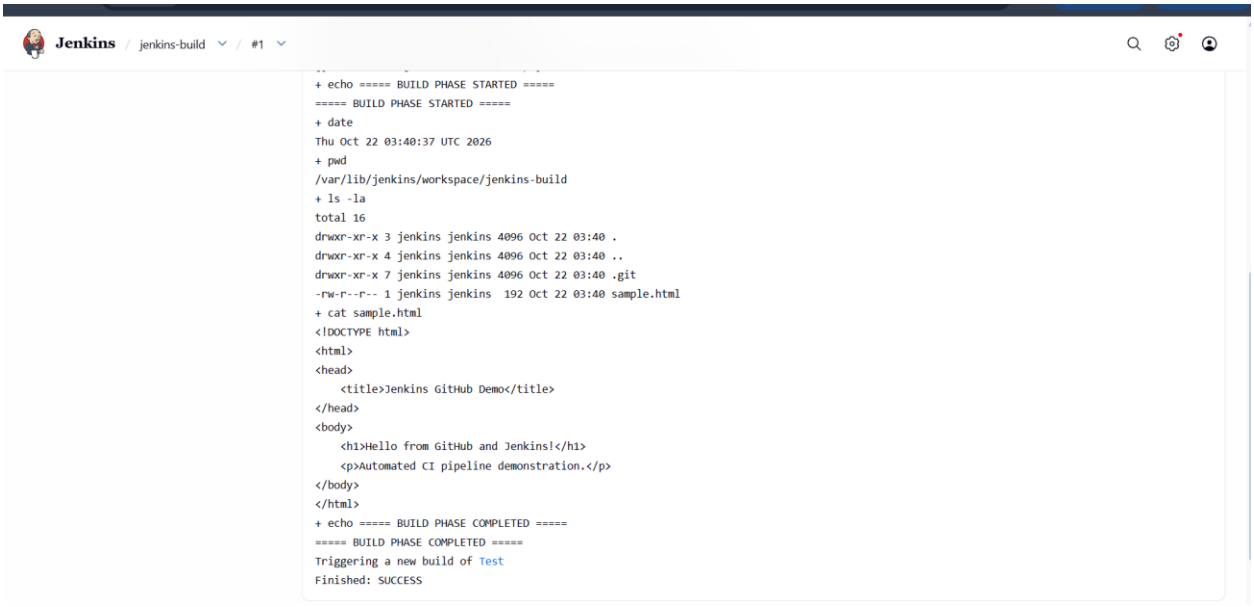


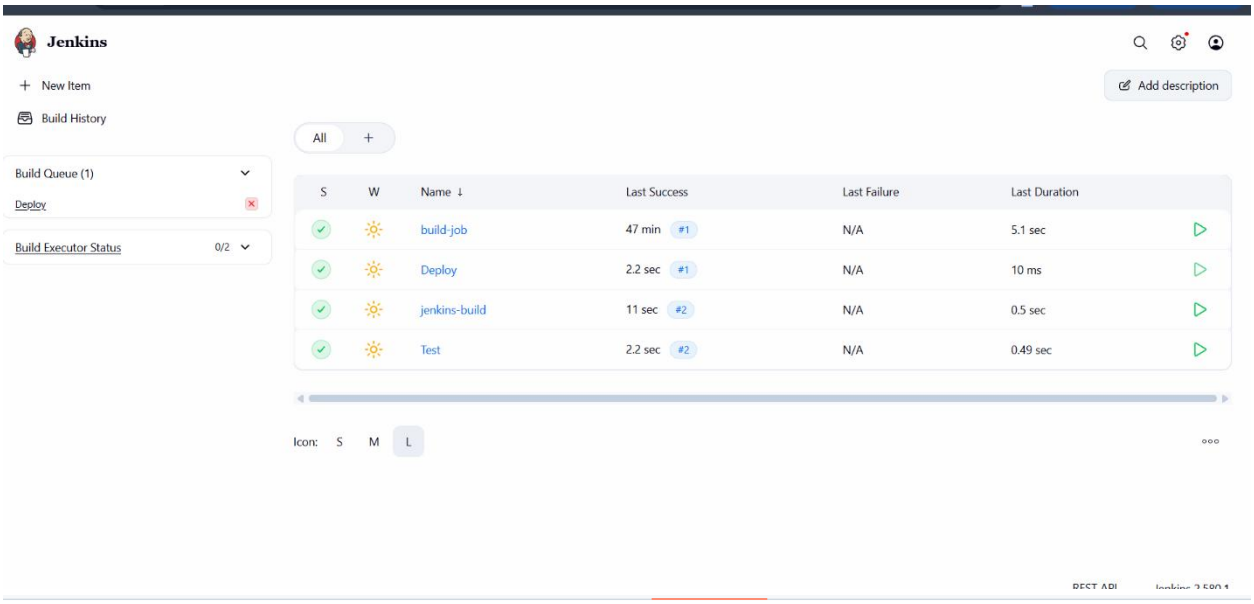

# GitHub and Jenkins Installation and Webhook Integration:




Not secure 13.60.55.140:8080/job/Deploy/1/console
Verify it's you
Action required
Jenkins / Deploy / #1
Status
Changes
Console Output
Delete build '#1'
Edit build information
Next Build
Console
Download
Copy
View as plain text
Started by upstream project "Test" build number 1
originally caused by:
Started by upstream project "jenkins-build" build number 1
originally caused by:
Started by user admin
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Deploy
Finished: SUCCESS


Not secure 13.60.55.140:8080/job/Test/1/console
Verify it's you
Action required
Jenkins / Test / #1
> git --version # timeout=10
> git --version # 'git version 2.53.0'
> git fetch --tags --force --progress -- https://github.com/SaiMeghana08/jenkins-github-demo.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/SaiMeghana08/jenkins-github-demo.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 3d13d0e82af3f9efae3d8b2ff66a9e600709f492 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 3d13d0e82af3f9efae3d8b2ff66a9e600709f492 # timeout=10
Commit message: "Index.html"
First time build. Skipping changelog.
[Test] $ /bin/sh -xe /tmp/jenkins534332741712914031 4.sh
+ echo ===== TEST PHASE STARTED =====
===== TEST PHASE STARTED =====
+ test -f sample.html
+ grep -q Hello from GitHub and Jenkins! sample.html
+ echo PASS: HTML file exists.
PASS: HTML file exists.
+ echo PASS: Expected text found.
PASS: Expected text found.
+ echo ===== TEST PHASE PASSED =====
===== TEST PHASE PASSED =====
Triggering a new build of Deploy
Finished: SUCCESS


Jenkins / jenkins-build / #1
+ echo ===== BUILD PHASE STARTED =====
===== BUILD PHASE STARTED =====
+ date
Thu Oct 22 03:40:37 UTC 2026
+ pwd
/var/lib/jenkins/workspace/jenkins-build
+ ls -la
total 16
drwxr-xr-x 3 jenkins jenkins 4096 Oct 22 03:40 .
drwxr-xr-x 4 jenkins jenkins 4096 Oct 22 03:40 ..
drwxr-xr-x 7 jenkins jenkins 4096 Oct 22 03:40 .git
-rw-r--r-- 1 jenkins jenkins 192 Oct 22 03:40 sample.html
+ cat sample.html
<!DOCTYPE html>
<html>
<head>
<title>Jenkins GitHub Demo</title>
</head>
<body>
<h1>Hello from GitHub and Jenkins!</h1>
Automated CI pipeline demonstration.
</body>
</html>
+ echo ===== BUILD PHASE COMPLETED =====
===== BUILD PHASE COMPLETED =====
Triggering a new build of Test
Finished: SUCCESS




# Webhooks

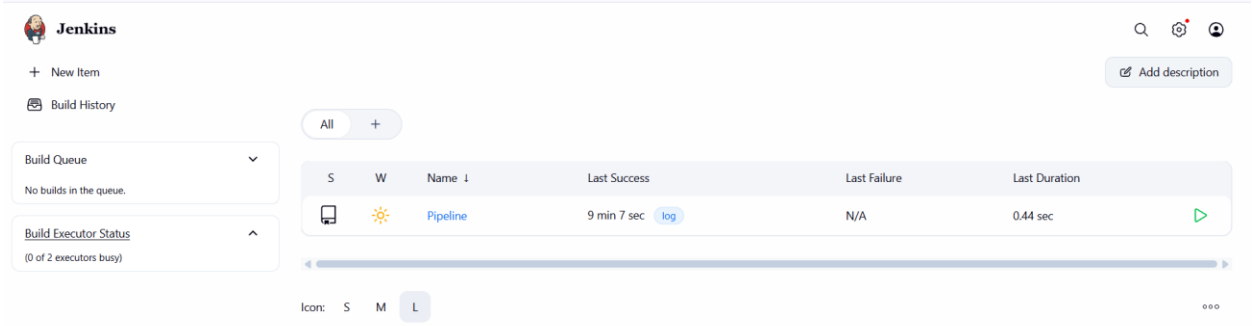
Add webhook

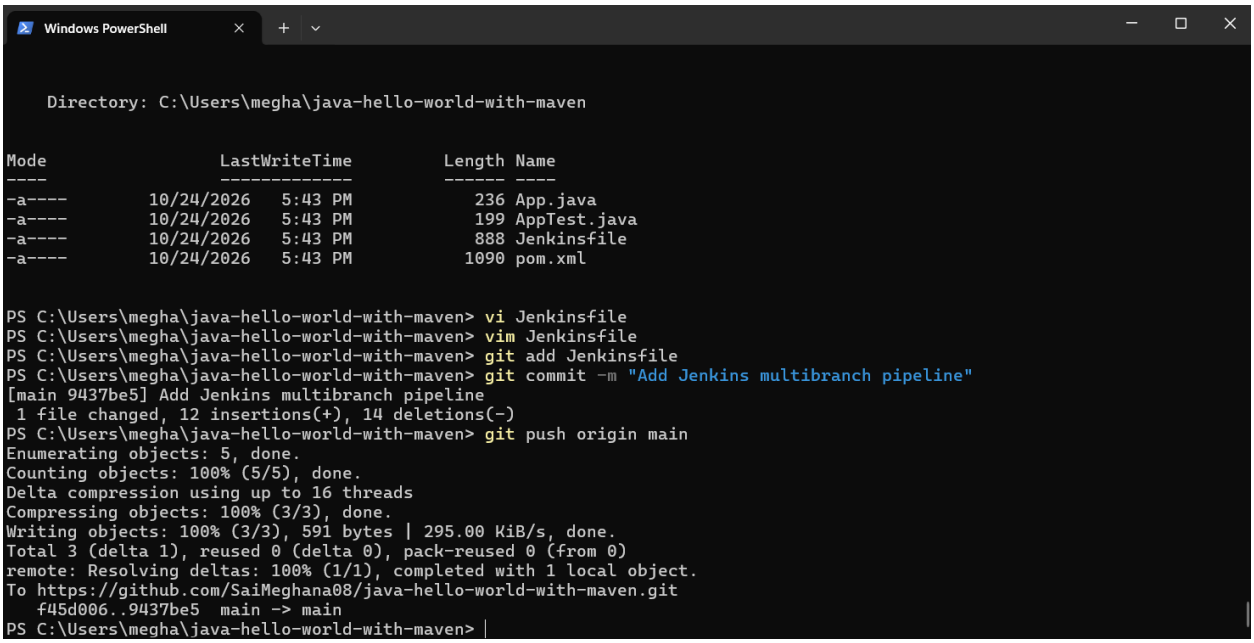
Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our Webhooks Guide.

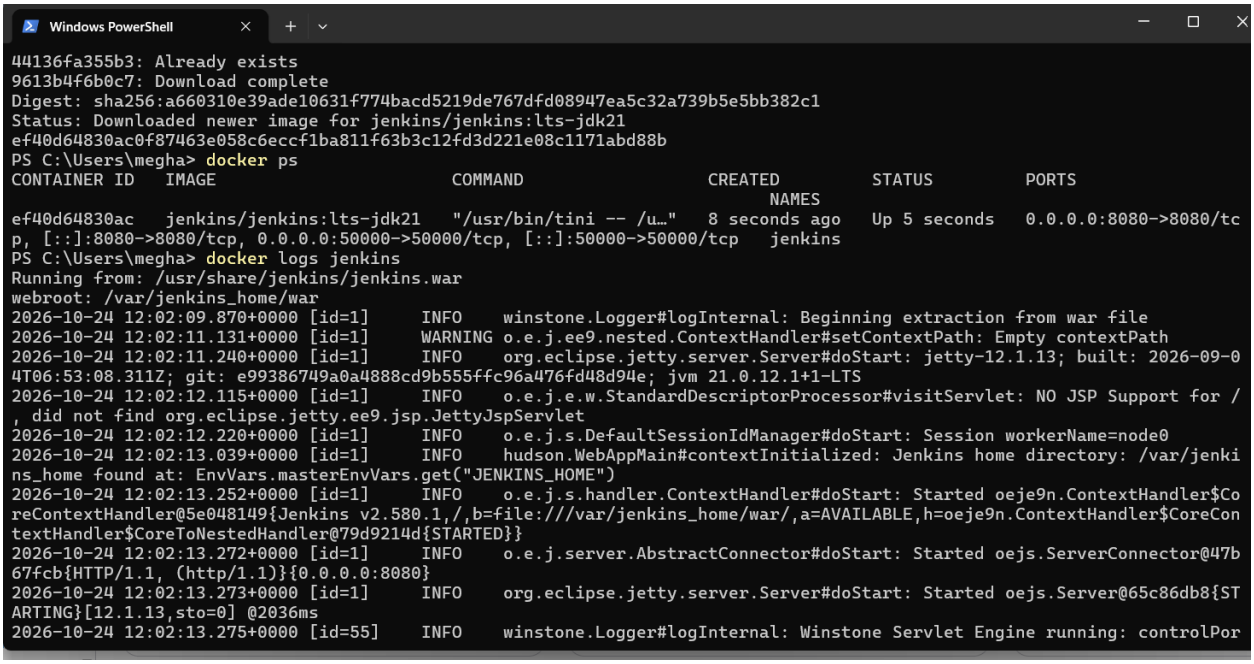
✓ http://13.60.55.140:8080/github-we... *(push)*

Edit Delete

Last delivery was successful.




Windows PowerShell
Directory: C:\Users\megha\java-hello-world-with-maven
Mode LastWriteTime Length Name
-a---- 10/24/2026 5:43 PM 236 App.java
-a---- 10/24/2026 5:43 PM 199 AppTest.java
-a---- 10/24/2026 5:43 PM 888 Jenkinsfile
-a---- 10/24/2026 5:43 PM 1090 pom.xml
PS C:\Users\megha\java-hello-world-with-maven> vi Jenkinsfile
PS C:\Users\megha\java-hello-world-with-maven> vim Jenkinsfile
PS C:\Users\megha\java-hello-world-with-maven> git add Jenkinsfile
PS C:\Users\megha\java-hello-world-with-maven> git commit -m "Add Jenkins multibranch pipeline"
[main 9437be5] Add Jenkins multibranch pipeline
1 file changed, 12 insertions(+), 14 deletions(-)
PS C:\Users\megha\java-hello-world-with-maven> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 591 bytes | 295.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/SaiMeghana08/java-hello-world-with-maven.git
f45d006..9437be5 main -> main
PS C:\Users\megha\java-hello-world-with-maven>


Windows PowerShell
44136fa355b3: Already exists
9613b4f6b0c7: Download complete
Digest: sha256:a660310e39ade10631f774bacd5219de767dfd08947ea5c32a739b5e5bb382c1
Status: Downloaded newer image for jenkins/jenkins:lts-jdk21
ef40d64830ac0f87463e058c6eccf1ba811f63b3c12fd3d221e08c1171abd88b
PS C:\Users\megha> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
ef40d64830ac jenkins/jenkins:lts-jdk21 "/usr/bin/tini -- /u…" 8 seconds ago Up 5 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp jenkins
PS C:\Users\megha> docker logs jenkins
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2026-10-24 12:02:09.870+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2026-10-24 12:02:11.131+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-10-24 12:02:11.240+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.1.13; built: 2026-09-04T06:53:08.311Z; git: e99386749a0a4888cd9b555ffc96a476fd48d94e; jvm 21.0.12.1+1-LTS
2026-10-24 12:02:12.115+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-10-24 12:02:12.220+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-10-24 12:02:13.039+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2026-10-24 12:02:13.252+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@5e048149{Jenkins v2.580.1,/,b=file:///var/jenkins_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@79d9214d{STARTED}}
2026-10-24 12:02:13.272+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@47b67fcb{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-10-24 12:02:13.273+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@65c86db8{STARTING}[12.1.13,sto=0] @2036ms
2026-10-24 12:02:13.275+0000 [id=55] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPor

```
indows PowerShell
opyright (C) Microsoft Corporation. All rights reserved.

S C:\Users\megha> docker --version
ocker version 29.4.1, build 055a478
S C:\Users\megha> docker ps
ailed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if th
 daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
S C:\Users\megha> docker ps
ONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
S C:\Users\megha> docker volume create jenkins_home
enkins_home
S C:\Users\megha> docker run -d --name jenkins --restart unless-stopped -p 8080:8080 -p 50000:50000 -v jenkins_home:/va
/jenkins_home jenkins/jenkins:lts-jdk21
nable to find image 'jenkins/jenkins:lts-jdk21' locally
ts-jdk21: Pulling from jenkins/jenkins
eefb2f5d3e9: Pull complete
91427fe0e3f: Pull complete
08777b27aa9: Pull complete
10008851f3a: Pull complete
89089c8a5a3: Pull complete
01ded2f9b33: Pull complete
68c36a3b2c5: Pull complete
dd4a99993ab: Pull complete
20a14499a3e: Pull complete
63a574793f8: Pull complete
9e31fab50e8: Pull complete
5bc7464aaea: Pull complete
4136fa355b3: Already exists
613b4f6b0c7: Download complete
```

